Guide for 'Your First Data Science Project' prerequisites

Thursday, October 23, 2025 2:16 PM

Setting Up Resource Principals and Policies

Scope

The following guide walks you through the minimal policies required to:

1. Create and use Notebook Sessions for running open-source libraries, registering models, deploying models, and reading/writing from Object Storage.
2. Register and deploy a model using AI Quick Actions.

Setup for Admin

As an Admin user, you already have user policies granting access to the relevant resources in OCI. Still, a few setup steps are required.

Creating a compartment

Compartments are logical containers where your resources reside. To use Data Science resources, you must have a compartment.

1. Go to Identity> Compartment
2. Create your compartment

Resources:

- OCI official documentation
- YouTube video

Dynamic Group

For resources to interact with each other, you need to define policies for resources.
To do this effectively, create a dynamic group that groups resources based on matching rules you define.
Policies assigned to the dynamic group automatically apply to all resources that match those rules.

1. Go to Identity> Domains>Dynamic groups
2. Click on Create Dynamic Group
3. Give it a name and add the following Rules:
   ALL {resource.type= 'datasciencenotebooksession'}
   ALL {resource.type='datasciencemodeldeployment'}

You may also want to add rules related to automation resources:
ALL { resource.type = 'datascienceschedule'}
ALL {resource.type='datasciencejobrun'}

Resources:

- OCI official documentation

Creating Dynamic Group Policies

1. Go to Identity> Policies
2. Create the following policies, and replace the placeholders with your actual names:

allow dynamic-group <your_dynamic_group> to manage data-science-family in compartment <your_compartment>
allow dynamic-group <your_dynamic_group> to manage object-family in compartment <your_compartment>
allow dynamic-group <your_dynamic_group> to use virtual-network-family in compartment <your_compartment>
allow service datascience to use virtual-network-family in compartment <your_compartment>
allow dynamic-group <your_dynamic_group> to read resource-availability in compartment <your_compartment>
allow dynamic-group <your_dynamic_group> to read repos in compartment <your_compartment>
allow dynamic-group <your_dynamic_group> to manage log-content in compartment <your_compartment>

Additional user group policies for a non-admin user

If you are not an Admin user, ask your Admin to add your group with the following policies:

allow group <your_group> to use virtual-network-family in compartment <your_compartment>

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allow group <your_group>  to manage data-science-family in compartment <your_compartment>
allow group <your_group>   to manage object-family in compartment <your_compartment>
allow group <your_group>   to manage tag-namespace in compartment <your_compartment>
allow group <your_group> to use logging-family in compartment <your_compartment>
```

Creating an Object Storage Bucket

1. Go to Storage> Buckets
2. Enable 'Create an object version when a new object is uploaded'.

Creating a Log Group and a Log (Optional)

1. Go to Observability & Management > Logging > Log Groups
2. Select 'Create log group'
3. Select the compartment and provide a log group name, and then 'Create'
4. Go to Observability & Management > Logging > Logs
5. Select 'Create custom log'
6. Add 'Custom log name', 'Resource compartment', 'Log group compartment', and 'Log group'
7. Click 'Next'
8. Select 'Add configuration later' and 'Next'
9. Review and Click 'Create'